Ahmedabad University

CSE 623 Machine Learning

Weekly Report - 5

Group- Vector Minds

Team

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Project Title: Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Project Definition: Agriculture plays an important role in our lives, especially in a well-known agricultural country like India. It provides food security to a larger population; however, its productivity depends heavily on factors such as rainfall and seasonal weather conditions. Due to increasing climate variability, predicting crop yields often becomes difficult.
The aim of this project is to develop a predictive system that estimates crop yield based on provided agricultural and historical data. The project applies a classical machine learning regression model to analyze the effect of environmental factors on agricultural productivity.

The project treats crop yield prediction as a supervised regression problem.

Where:

- Input variables: State, Crop type, Season, Crop Year, Area, Annual Rainfall, Fertilizer, Pesticide, and four engineered features — Fertilizer per Area, Pesticide per Area, Rainfall per Area, and Input Intensity
- Output variable (Target): Crop yield, which is production per unit area.

Objectives for Week 5

1. Finding the optimal model configuration efficiently with the help of hyperparameter tuning on a Random Forest regressor using RandomizedSearchCV
2. Implementation of advanced versions of Gradient Boosting for better and accurate results.
3. Performs error distribution of the best model to assess the symmetry of its predictions.
4. Generate visualizations and comparative evaluations across all models from overall project conclusions.

5. Focus more on feature engineering and importance by adding new parameters from existing ones.

Work Completed

During this week, the project focused on model comparison, model optimization and ensemble model construction. The best performing models were combined into a stacking ensemble and later evaluated to produce the final prediction system.

Data Quality & Feature Engineering

Before the model training, quality improvements are being applied to the dataset. For the right-skewed distributions in fertilizer, pesticide and area, we corrected them using lop1 p transformation. IQR-based capping is applied to handle outliers and preserve the dataset size. After all this, new engineered features like fertilizer per unit area, pesticide per unit area, rainfall per unit area and input intensity(combined fertilizer and Pesticide) were made using the existing ones. EDA provides a comparison between the two, before and after transformations, to assess improvements in data distribution.

Hyperparameter Tuning- Random Forest (RandomizedSearchCV)

RandomizedSearchCV was applied to the Random Forest pipeline, with 30 random parameter combinations sampled from their distributions, using 3-fold cross-validation with R^2 as the scoring metric. RandomizedSearchCV was preferred over GridSearchCV because the large parameter space, with random sampling, provides a strong near-optimal solution in significantly less computation time than a grid search.

Gradient Boosting Model- Gradient Boosting (GridSearchCV)

Later, GridSearchCV was applied on the Gradient Boosting pipeline to evaluate the combinations of n_estimators (100, 200), learning_rate (0.05, 0.1, 0.2), max_depth (3, 5, 7), and subsample (0.8, 1.0) using 3-fold cross-validation with R^2 scoring. GridSearchCV was chosen for Gradient Boosting as its parameter space was smaller and structured, making an exhaustive search feasible and providing the optimal combination without missing any parameters.

Stacking Ensemble Model

A stacking regressor was constructed with three different base learners: Random Forest (n_estimators=150, max_depth=15), Gradient Boosting (n_estimators=150, learning_rate=0.1, max_depth=5) and Ridge Regression (alpha=1.0). Ridge Regression (alpha=0.5) was used as the meta-learner. The stacking model was trained using 5-fold cross-validation to generate predictions from the base learning models. The design ensures that meta-learners optimally combine the strengths of each base model. The preprocessor was applied before training the stacking model, since it operates directly on the transformed feature matrices.

Cross-Validation

To evaluate the models' reliability, 5-fold cross-validation was performed on the RF Baseline. Tuned Random Forest and Tuned Gradient Boosting compare the generalization performance of two models after tuning. The mean R^2 and standard deviation across all folds were reported for each model to confirm that tuning improved test-set performance and also cross-validated stability.

Model Evaluation

Each of the tuned models and the ensemble model were evaluated on the same test data using:

- Mean Absolute Error (MAE) – It calculates the average absolute prediction error.
- Root Mean Squared Error (RMSE) – It gives more weight to larger errors.
- R^2 Score – It calculates the percentage of yield that the model explains.

Model Visualization

A learning curve plot has been created for the Tuned Random Forest model, showing the training and validation R^2 values as a function of the training set size. A comparison bar chart has been created to compare the performance of the final models based on MAE, RMSE, and R^2 values, including the four models used in Week 4: RF Baseline, Tuned Random Forest, Tuned Gradient Boosting, and Stacking Ensemble.

A feature importance bar chart has been created for the Tuned Random Forest model to determine whether the tuned model has the same feature importance as the default model developed in Week 3.

An actual vs. predicted scatter plot has been created for the best model to check the quality of the predictions.

A histogram of the prediction error distribution with a kernel density estimate has been created to examine the distribution of prediction errors.

If we compare distribution plots before and after transformation, the correlation hetmap gives distribution plots for all engineered features and boxplots are generated to verify the improvements in data quality.

Final Model Comparison

A final model comparison was performed based on the MAE, RMSE, and R^2 metrics. The results were presented by means of comparative charts to determine which model performed best. The results revealed that the Tuned Random Forest model had excellent prediction results and was determined to be the best-performing model for this dataset.

Final Observations

The implementation of ensemble models and hyperparameter tuning improved the accuracy of the crop yield prediction system. The tree-based ensemble model, Random Forest, and Gradient Boosting algorithms were highly effective for this problem. Furthermore, visualization and feature engineering analysis provided insights into the effect of climatic and agricultural factors on crop production. These factors are critical to understanding the complex relationships involved in agriculture.